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Red Team Lab

Purpose

This repository documents my journey toward becoming a professional Red Team Operator through structured, hands-on practice.

The focus is not simply on learning tools or completing labs, but on understanding the reasoning behind every action, when techniques should be used, their operational security (OPSEC) implications, possible detection opportunities, and how findings would be documented and reported during a real engagement.

My current focus is **local Windows post-exploitation**. Future phases will expand into multi-host environments, Active Directory, and enterprise-style operations.

Learning Philosophy

This repository is built around a "**learn by doing**" approach.

For every technique I study, I aim to understand:

- Why the technique exists.
- What problem it solves.
- When it should and should not be used.
- What artifacts it creates.
- How defenders can detect it.
- What operational risks it introduces.
- How it would be documented during an engagement.

Windows Internals is used as a reference when deeper understanding is required rather than something to study cover to cover before practising.

Folder Structure

00 - Lab Setup

Contains documentation for building and maintaining the lab environment, virtual machines, configurations, snapshots, and tooling.

01 - Foundations

High-level concepts that provide the knowledge required before performing post-exploitation activities.

02 - Windows Internals (Reference)

Reference notes on Windows architecture, authentication, services, processes, tokens, memory, and other internals studied as needed to support practical techniques.

03 - Local Enumeration

Host enumeration performed after obtaining code execution on a Windows system.

04 - Privilege Escalation

Techniques used to move from a lower-privileged account to higher privileges.

05 - Credential Access

Research and practice related to discovering, extracting, and understanding credentials and authentication material.

06 - Persistence

Methods used to maintain access across reboots or future logins.

07 - Defence Evasion

Techniques that reduce the likelihood of detection or interfere with defensive controls.

08 - Lateral Movement

Techniques used to move between systems after compromising an environment. This section will become relevant once the lab expands beyond a single machine.

09 - Collection

Methods for identifying and gathering valuable information from compromised systems.

10 - Exfiltration

Concepts and techniques related to moving collected data out of an environment.

11 - Clean-up

Removing or minimizing operational artifacts where appropriate and documenting what should be cleaned after an engagement.

12 - Reporting

Documentation, findings, timelines, screenshots, evidence, and engagement-style reports.

Cheat-sheets

Frequently used commands, workflows, syntax, and quick-reference notes.

References

Books, blog posts, white-papers, Microsoft documentation, conference talks, and other learning resources.

Labs

Hands-on exercises, walk-throughs, experiments, and personal notes from practical sessions.

Long-Term Goal

The objective of this repository is to build the technical knowledge, operational thinking, documentation habits, and disciplined methodology expected of a professional Red Team operator by progressively mastering Windows post-exploitation before expanding into enterprise environments, Active Directory, and full adversary emulation.